

## Exercises for MATH-4100

Fall 2025

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- 1) Given two vectors  $\mathbf{a} = (1, 1, 2)$  and  $\mathbf{b} = (3, 0, -1)$ , find a third vector  $\mathbf{c} \in \mathbb{R}^3$  that orthogonal to both  $\mathbf{a}$  and  $\mathbf{b}$ .
- 2) Let  $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$  be such that  $\mathbf{a} + \lambda \mathbf{b} \neq \mathbf{0}$  for all  $\lambda \in \mathbb{R}$ . Show that if both  $\mathbf{c}$  and  $\mathbf{d}$  are othogonal to  $\mathbf{a}$  and  $\mathbf{b}$  then  $\mathbf{c}$  and  $\mathbf{d}$  must be parallel.
- 3) Prove the parallelogram law:  $\|\mathbf{x} + \mathbf{y}\|^2 + \|\mathbf{x} - \mathbf{y}\|^2 = 2(\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2)$  for any  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ .
- 4) Let  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ . Show that:
  - a.  $\mathbf{x}^T \mathbf{y} = 0$  if and only if  $\|\mathbf{x}\| \leq \|\mathbf{x} + a\mathbf{y}\|$  for all  $a \in \mathbb{R}$
  - b. If  $\|\mathbf{x}\| = \|\mathbf{y}\|$  then  $\|a\mathbf{x} + b\mathbf{y}\| = \|b\mathbf{x} + a\mathbf{y}\|$  for all  $a, b \in \mathbb{R}$

- 5) Let  $a, b$ , and  $c$  be lenghts of the sides of a triangle and let  $d$  be the length of the line segment from the midpoint of the side of length  $c$  to the opposite vertex. Use inner products to prove that

$$a^2 + b^2 = \frac{1}{2}c^2 + 2d^2$$

- 6) Let  $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$  be a set of vectors in  $\mathbb{R}^n$  such that  $\mathbf{x}_i^T \mathbf{x}_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$ . Show that if  $\mathbf{v} = \sum c_i \mathbf{x}_i$  then  $c_i = \mathbf{x}_i^T \mathbf{v}$ . (Note that the vectors  $\mathbf{x}_i$  are said to be orthonormal)

- 7) Let  $V$  be an inner product space and let  $\{e_1, \dots, e_n\}$  be an orthonormal basis for  $V$ , that is  $\langle e_i, e_j \rangle = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$ . Show that for any  $x, y \in V$  with  $x = \sum_{i=1}^n x_i e_i$  and  $y = \sum_{i=1}^n y_i e_i$  we have that  $\langle x, y \rangle = \sum_{i=1}^n x_i y_i$ .

- 8) Show that the following subsets of  $\mathbb{R}^n$  are subspaces:

- a.  $W_1 = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} = \sum c_i \mathbf{v}_i\}$  where  $\mathbf{v}_i \in \mathbb{R}^n$  for each  $i$ . (This is the set of linear combinations of the vectors  $\mathbf{v}_i$ )
- b.  $W_2 = \{\mathbf{y} \in \mathbb{R}^n : \mathbf{y} = A\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^n\}$  where  $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$ .

- 9) Show that the following subsets of  $\mathbb{R}^n$  are not subspaces:

- a.  $S = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{b}\}$  where  $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$  and  $\mathbf{b} \neq \mathbf{0} \in \mathbb{R}^n$ .
- b. Any non-empty, finite subset of  $\mathbb{R}^n$ . (Clarification: by non-empty I mean containing at least one non-zero vector)

- 10) Let  $V$  be a vector space over  $F$  and let  $U_1$  and  $U_2$  be subspaces. Prove that  $U_1 \cap U_2$  is also a subspace.

- 11) Show by example that the union of two subspaces need not be a subspace.

- 12) Let  $V$  be a vector space over  $F$  and let  $S$  and  $T$  be subspaces such that  $U_1 \cap U_2 = \emptyset$ . Show that for any  $x \in S + T$ , there exists unique  $s \in S$  and  $t \in T$  such that  $x = s + t$ .
- 13) Given  $U_1 = \{(x_1, 0, x_3) \in \mathbb{R}^3 : 4x_1 + 2x_3 = 0\}$  and  $U_2 = \{(0, x_2, x_3) \in \mathbb{R}^3 : x_2 = 5x_3\}$ , compute  $U_1 + U_2$ .
- 14) Let  $W_1 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 = 3x_2\}$  and  $W_2 = \{(t, -3t, 0) : t \in \mathbb{R}\}$  be two subspaces of  $\mathbb{R}^3$ . Show that  $\mathbb{R}^3 = W_1 \oplus W_2$ .
- 15) Determine whether or not the following three vectors in  $\mathbb{R}^3$  linearly independent and describe their span

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix}, v_3 = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$$

- 16) Let  $V$  be a finite dimensional vector space and let  $\mathcal{B}$  and  $\mathcal{B}'$  be two bases. Show that  $|\mathcal{B}| = |\mathcal{B}'|$ , i.e. show that both sets have the same number of elements.
- 17) Let  $\mathbb{P}_n$  be the space of all polynomials over  $\mathbb{R}$  of degree at most  $n$ . Show that the subset  $S = \{f \in \mathbb{P}_n : f'(1) = 0\}$  is a subspace of  $\mathbb{P}_n$  and find a basis.
- 18) Let  $V$  be a vector space and let  $L$  be any linear operator on  $V$ , i.e.  $L \in \mathcal{L}(V, V)$ . Prove that the subset  $S = \{y \in V : y = L[x] \text{ for some } x \in V\}$  is a subspace of  $V$ .
- 19) Let  $T \in \mathcal{L}(V, W)$  be a linear transform and define  $\ker(T) = \{x \in V : T(x) = \mathbf{0}\}$ . Prove that  $T$  is injective if and only if  $\ker(T) = \mathbf{0}$ . ( $\ker$ =kernel)
- 20) Let  $V$  be finite dimensional and let  $T \in \mathcal{L}(V, V)$ . Show that  $T$  is surjective if and only if  $\ker(T) = \mathbf{0}$ . Does this remain true if  $V$  is infinite dimensional?

21) Given  $A = \begin{bmatrix} 1 & 2 & 4 \\ -2 & -3 & 2 \\ 2 & 2 & -1 \end{bmatrix}$ , and  $B = \begin{bmatrix} 3 & -2 & 6 \\ 1 & 1 & 0 \\ -1 & 3 & 2 \end{bmatrix}$  compute  $AB$  and  $BA$

- 22) Find the null space and range of the following matrices

a.  $A = \begin{bmatrix} 1 & 2 & 0 \\ 6 & 5 & 1 \\ -3 & 1 & -1 \end{bmatrix}$

b.  $A = \begin{bmatrix} 1 & -2 & 1 & 0 & 3 \\ -3 & 6 & 0 & 2 & -1 \\ -2 & 4 & 3 & 1 & 0 \end{bmatrix}$

- 23)  $A \in \mathbb{R}^{m \times n}$  and  $B \in \mathbb{R}^{n \times l}$ . Prove that  $\text{null}(B) \subseteq \text{null}(AB)$  and  $\text{range}(AB) \subseteq \text{range}(A)$ . (This demonstrates that  $\text{nullity}(B) \leq \text{nullity}(AB)$  and  $\text{rank}(AB) \leq \text{rank}(A)$ .)
- 24) Let  $A$  and  $B$  be invertible  $n \times n$  matrices. Prove that
- a.  $(AB)^{-1} = B^{-1}A^{-1}$
- b.  $(A^{-1})^T = (A^T)^{-1}$

- 25) Given  $A = \begin{bmatrix} 1 & 2 & 4 \\ -2 & 0 & 2 \\ 2 & 2 & -1 \end{bmatrix}$ , compute the following
- $A^2$
  - $A^T$
  - $A^{-1}$
- 26) Prove that if  $A \in \mathbb{R}^{m \times n}$  where  $n \neq m$ , then  $A$  is not invertible.
- 27) Let  $A \in \mathbb{R}^{m \times n}$  where  $m < n$  be such that  $\text{rank}(A) = m$ . Show that  $AA^T$  is invertible, but  $A^T A$  is not.
- 28) Show that every matrix can be decomposed into a symmetric and skew symmetric matrix. I.e, for any matrix  $A$  we can find matrices  $B$  and  $C$  such that  $B^T = B$ ,  $C^T = -C$ , and  $A = B + C$ . (This corresponds with the property that any function can be written as the sum of an even and odd function.)
- 29) Find the general solution to  $A\mathbf{x} = \mathbf{b}$  given that
- $$A = \begin{bmatrix} 1 & 0 & 2 & 4 \\ 2 & 1 & 1 & 3 \\ 3 & 1 & 3 & 7 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 2 \\ 6 \\ 8 \end{bmatrix}$$
- 30) Let  $S$  be the subspace of  $\mathbb{R}^4$  given by  $S = \{(x_1, x_2, x_3, x_4) : x_1 - 2x_2 + 4x_3 - 3x_4 = 0\}$
- Find a basis for  $S$  and  $S^\perp$
  - Find a matrix  $M$  such that  $S^\perp = \text{null}(M)$
- 31) Let  $S$  be the subspace of  $\mathbb{P}_3$  given by  $S = \{f \in \mathbb{P}_3 : f(1) = f'(1) = 0\}$ . Find bases for  $S$  and  $S^\perp$ . (You may assume an inner product where the standard basis is orthonormal)
- 32) Find a basis for  $S^\perp$  from above using the typical inner product,  $\langle f, g \rangle = \int_0^1 fg$
- 33) Prove that given any matrix  $A$ ,  $\text{range}(A)^\perp = \text{null}(A^T)$  and  $\text{range}(A^T)^\perp = \text{null}(A)$ .
- 34) Compute  $\text{range}(A^T)$  and  $\text{null}(A^T)$
- $A = \begin{bmatrix} 1 & 2 & 0 \\ 6 & 5 & 1 \\ -3 & 1 & -1 \end{bmatrix}$
  - $A = \begin{bmatrix} 1 & -2 & 1 & 0 & 3 \\ -3 & 6 & 0 & 2 & -1 \\ -2 & 4 & 3 & 1 & 0 \end{bmatrix}$
- 35) Let  $A \in \mathbb{R}^{m \times n}$  and  $B \in \mathbb{R}^{m \times l}$ . Show that  $\text{range}(A)$  is orthogonal to  $\text{range}(B)$  if and only if  $A^T B$  is the zero matrix
- 36) Recall that a directed graph consists of nodes and directed edges connecting two nodes. We define a connected component of the graph to be a collection of nodes that share an edge with at least one other node in the collection. Let  $A$  be the incidence matrix for the directed graph  $G$ . Show that  $\text{nullity}(A)$  is equal to the number of connected components of  $G$ .

37) Let

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 \end{bmatrix}$$

- Draw a directed graph having  $A$  as its incidence matrix
- Find bases for the four fundamental subspaces;  $\text{range}(A)$ ,  $\text{range}(A^T)$ ,  $\text{null}(A)$ , and  $\text{null}(A^T)$

38) Let  $\{\mathbf{e}_i\}$  be the standard basis for  $\mathbb{R}^2$ . Let  $T$  be the linear transformation on  $\mathbb{R}^2$  defined as follows

$$\begin{aligned} T\mathbf{e}_1 &= \mathbf{e}_1 + 2\mathbf{e}_2 \\ T\mathbf{e}_2 &= 3\mathbf{e}_1 - 3\mathbf{e}_2 \end{aligned}$$

- Find  $[T]_{\{\mathbf{e}_i\}}$  the matrix representation of  $T$  with respect to the standard basis.
- Let  $\{\mathbf{v}_1, \mathbf{v}_2\}$  be another basis for  $\mathbb{R}^2$  where  $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$  and  $\mathbf{v}_2 = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$ . Find  $[T]_{\{\mathbf{v}_i\}}$ , the matrix representation of  $T$  with respect to the basis  $\{\mathbf{v}_i\}$  and use it write  $T \begin{bmatrix} 4 \\ 5 \end{bmatrix}$  as a linear combination of the basis vectors,  $\mathbf{v}_i$ .

39) Let  $T$  be a linear transformation on  $\mathbb{R}^n$  such that  $T\mathbf{e}_i = \mathbf{v}_i$  for  $i = 1, \dots, n$ .

- Find  $[T]_{\{\mathbf{e}_i\}}$  the matrix representation of  $T$  w.r.t the standard basis
- Now assume that for linearly independent vectors  $\{\mathbf{w}_1, \dots, \mathbf{w}_n\}$  we have that  $T\mathbf{w}_i = \lambda_i \mathbf{w}_i$  for each  $i$ . Show that the matrix representation of  $[T]_{\{\mathbf{w}_i\}}$  is a diagonal matrix.
- Finally state the change of basis matrix from  $\{\mathbf{e}_i\}$  to  $\{\mathbf{w}_i\}$  and find a relationship between each of the previously mentioned matrices.

40) Let  $L : \mathbb{P}_3 \rightarrow \mathbb{P}_3$  be defined by  $L[f] = D[(2x + 3)f]$  where  $D = d/dx$  is the differential operator.

- Show that  $L$  is a linear operator
- Find  $[L]$  with respect to the standard basis  $\{1, x, x^2, x^3\}$  and show that  $L$  is onto. (Think dimensionality)
- Use the above result to solve the equation  $L[f] = x^2$ . You may use a CAS here.

41) Let  $T : \mathbb{P}_2 \rightarrow \mathbb{R}_3$  be defined by  $T(f) = (f(-3), f(2), f(5))$ .

- Show that  $T$  is a linear operator
- Find  $[T]$  with respect to the standard basis  $\{1, x, x^2\}$  and show that  $\ker(T) = \{0\}$
- Use the above result to find the equation of the parabola passing through the points,  $(-3, 6)$ ,  $(2, -3)$ ,  $(5, 4)$

42) Let  $A$  be any matrix and prove that

- $\text{null}(A^T A) = \text{null}(A)$

b.  $\text{null}(AA^T) = \text{null}(A^T)$

Hint: Use the orthogonality of the fundamental subspaces. To prove set equality show that they are both subsets of each other:  $S \subseteq T$  iff  $x \in S \Rightarrow x \in T$

43) Let  $P$  be a (not necessarily orthogonal) projection.

- Prove that for any  $\mathbf{y} \in \text{range}(P)$ ,  $P\mathbf{y} = \mathbf{y}$ .
- Prove that  $\text{range}(P) = \text{null}(I - P)$ .
- Prove that if  $P$  is full rank, then it must be the identity.

44) Let  $V$  be a finite dimensional vector space and let  $P$  be an orthogonal projection onto the subspace  $W$ . Show that  $\|P\mathbf{x}\| \leq \|\mathbf{x}\|$  for all  $\mathbf{x} \in V$ . (Such a map is called a contraction)

45) Prove the converse, that is if  $P$  is a projection onto  $W$  that is also a contraction, then  $P$  is an orthogonal projection onto  $W$ .

46) Let  $S = \text{span}(\mathbf{v}_1, \mathbf{v}_2)$  where  $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \\ 4 \\ -1 \end{bmatrix}$ , and  $\mathbf{v}_2 = \begin{bmatrix} 3 \\ 0 \\ -1 \\ 1 \end{bmatrix}$ . Find the projection matrix onto  $S$  and use it to find the vector in  $S$  that is closest to  $(4, 1, 2, -2)$ .

47) Use linear algebra to find the line of best fit given the set of points  $\{(-1, 11), (0, 4), (1, -2), (2, 1), (3, -6)\}$

48) Given vectors,  $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ ,  $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$ ,  $\mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$ , use Gram-Schmidt to find a corresponding orthonormal set  $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\}$  and show that  $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\} = \text{Span}\{\mathbf{q}_1, \mathbf{q}_2\}$ .

49) Assume that  $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$  are mutually orthogonal and show that for any  $\mathbf{v}$ ,  $\mathbf{v} - \sum_{i=1}^n \frac{\mathbf{u}_i^T \mathbf{v}}{\mathbf{u}_i^T \mathbf{u}_i} \mathbf{u}_i$  is orthogonal to each  $\mathbf{u}_i$ .

50) Suppose that  $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$  is an orthonormal basis of  $\mathbb{R}^n$  and that  $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$  are such that

$$\|\mathbf{e}_i - \mathbf{v}_i\| < 1/\sqrt{n}$$

for each  $i$ . Show that  $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$  is also a basis of  $\mathbb{R}^n$ .

51) Let

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 4 & 0 \\ 4 & -7 & 9 \end{bmatrix}$$

find the  $LU$  factorization for  $A$  and use it to solve the system

$$\begin{aligned} 2x_1 + x_2 + x_3 &= 6 \\ 2x_1 + 4x_2 &= 0 \\ 4x_1 - 7x_2 + 9x_3 &= 42 \end{aligned}$$

52) Find the reduced  $QR$  factorization of the matrix

$$A = \begin{bmatrix} 1 & -2 \\ -1 & 1 \\ 3 & 1 \end{bmatrix}$$

53) Let  $A \in \mathbb{R}^{2 \times 2}$  be such that for distinct  $\lambda_1, \lambda_2 \in \mathbb{R}$ ,  $A - \lambda_1 I$  and  $A - \lambda_2 I$  are not the zero matrix. Prove that if  $A^2 - (\lambda_1 + \lambda_2)A + \lambda_1 \lambda_2 I = \mathbf{0}$ , then  $A - \lambda_1 I$  and  $A - \lambda_2 I$  are not invertible and that the column vectors of  $A - \lambda_1 I$  are contained in the  $\text{null}(A - \lambda_2 I)$ .

54) Find  $\det A$  (by hand) using any method

$$A = \begin{bmatrix} 2 & 4 & 0 & 3 \\ -1 & 3 & 1 & 1 \\ 0 & 0 & 3 & 0 \\ -3 & 0 & 1 & 2 \end{bmatrix}$$

55) Use Cramer's rule to solve the system

$$\begin{aligned} 2x_1 + 2x_3 &= 2 \\ x_2 - 3x_3 &= 5 \\ 2x_1 - x_2 + x_3 &= 5 \end{aligned}$$

56) Given  $A = \begin{bmatrix} 2 & 0 & 1 \\ -3 & 2 & 4 \\ 1 & -2 & 4 \end{bmatrix}$  compute  $\text{adj}(A)$  and use it to find  $A^{-1}$ .

57) Prove that for  $A \in \mathbb{R}^{n \times n}$ ,  $\text{rank}(\text{adj}(A)) = n - \text{rank}(A)$

58) Let  $P$  be a projection matrix and  $Q$  be an orthogonal matrix. Prove that  $|P|$  must be either 1 or 0 and  $|Q|$  must be either  $\pm 1$ .

59) Let  $H_n$  be the  $n \times n$  Hilbert matrix defined by,  $H_{ij} = \frac{1}{i+j-1}$ . Compute  $|H_2|$  and  $|H_3|$  and note that the determinant is the reciprocal of an integer and that the components of the inverse matrices become quite large. This means that these matrices are ill-conditioned and therefore are not well suited for computation.

60) Find the eigenvalues and eigenvectors given the following matrices:

a.  $A = \begin{bmatrix} 2 & 9 \\ -2 & -7 \end{bmatrix}$

b.  $B = \begin{bmatrix} 3 & 2 & -1 \\ 0 & 0 & 1 \\ 0 & -4 & 3 \end{bmatrix}$

61) Find the minimal polynomial of the matrix  $A = \begin{bmatrix} 9 & 9 \\ -4 & -3 \end{bmatrix}$

62) Find the set of all  $A \in \mathbb{R}^{2 \times 2}$  such that  $A^8 = I$  up to similarity classes. I.e. find all possible minimal polynomials.

- 63) Let  $A$  be a  $2 \times 2$  integer matrix such that  $A^k = I$  for some  $k \in \mathbb{N}$ . Show that  $A^{12} = I$ . **Hint:** A result in HW 6 may be helpful.
- 64) Prove that if  $A$  is its own inverse, then the eigenvalues of  $A$  are either 1 or -1
- 65) Let  $A$  is invertible if and only if the constant term of the minimal polynomial is nonzero.
- 66) Let  $T \in \mathcal{L}(\mathbb{C}^n, \mathbb{C}^n)$  be an isometry, i.e.  $\|T\mathbf{x}\| = \|\mathbf{x}\|$  for all  $\mathbf{x}$ , and let  $A$  be a matrix representation. Show that the constant term of the characteristic polynomial of  $A$  has absolute value 1.
- 67) Let  $P$  be a projection onto the subspace  $W$  of  $\mathbb{R}^n$  such that  $\dim W = r$ . State the characteristic polynomial of  $P$ . **Hint:** First find the minimal polynomial and reason that the zeros are eigenvalues. Next find the multiplicity of each eigenvalue by using the definition of  $P$ .
- 68) Let  $M$  be the Markov matrix

$$M = \begin{bmatrix} 0.4 & 0.9 \\ 0.6 & 0.1 \end{bmatrix}.$$

Find the diagonalization of  $M$  and use it to find the steady-state matrix  $M^\infty = \lim M^k$ .

- 69) Suppose that  $(s_n)$  is the linear-recursive sequence defined by

$$s_{n+1} = 3s_n - 2s_{n-1}, \quad s_0 = 0, \quad s_1 = 1$$

using the method outlined in the lecture, define an appropriate difference equation and use it to find a closed form for  $s_n$ . Then find the limit of the sequence  $(t_n)$  where  $t_n = s_{n+1}/s_n$  for  $n \geq 1$ .

- 70) Find the (complex) diagonalization of the matrix  $A = \begin{bmatrix} 4 & -1 \\ 5 & 2 \end{bmatrix}$

- 71) Use linear algebra to solve the system of differential equations

$$\begin{aligned} x_1' &= -x_1 + 4x_2 \\ x_2' &= 4x_1 + 5x_2 \end{aligned}$$

with initial values  $x_1(0) = -2$ ,  $x_2(0) = 6$

- 72) Find the general solution to the non-homogeneous system

$$\mathbf{x}' = \begin{bmatrix} -1 & 4 \\ 4 & 5 \end{bmatrix} \mathbf{x} + \begin{bmatrix} e^t \\ e^{2t} \end{bmatrix}$$

- 73) Recall that for square matrix  $A$ , the matrix exponential  $e^{At}$  is defined by the formal power series

$$e^{At} = \sum_{n=0}^{\infty} \frac{A^n t^n}{n!}$$

Where  $A^0 = I$ . Prove the following results related to the matrix exponential:

- If  $A = SAS^{-1}$ , then  $e^{At} = Se^{\Lambda t}S^{-1}$
- If  $AB = BA$ , then  $e^{At}e^{Bt} = e^{(A+B)t}$
- $(e^{At})^{-1} = e^{A(-t)}$
- If  $P$  is a projection, then  $e^{Pt} = I + P(e^t - 1)$